



The present state and challenges of active learning in drug discovery

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Active learning (AL) is an iterative feedback process that efficiently identifies valuable data within vast chemical space, even with limited labeled data. This characteristic renders it a valuable approach to tackle the ongoing challenges faced in drug discovery, such as the ever-expanding explore space and the limitations of labeled data. Consequently, AL is increasingly gaining prominence in the field of drug development. In this paper, we comprehensively review the application of AL at all stages of drug discovery, including compounds–target interaction prediction, virtual screening, molecular generation and optimization, as well as molecular properties prediction. Additionally, we discuss the challenges and prospects associated with the current applications of AL in drug discovery.

Keywords: active learning; compounds–target interaction prediction; virtual screening; molecular generation and optimization; molecular properties prediction

Introduction

The primary objective of drug discovery is to pinpoint specific target molecules possessing desirable characteristics, such as biological activity and druglike properties, within the vast chemical space. However, the rapid expansion of the chemical space has

made the traditional approach of identifying target molecules through experimentation impractical. Integrating machine learning (ML) algorithms into the field of drug discovery can offer valuable guidance for navigating this complex chemical space, thereby expediting the drug discovery process.^{(p1),(p2)} Nevertheless, the



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effective application of ML is hindered by the limited availability of labeled data and the resource-intensive nature of obtaining such data.^(p3) Furthermore, challenges such as data imbalance and redundancy within labeled datasets also impede the application of ML.^{(p4),(p5)} In this context, active learning (AL) algorithms emerged as a compelling solution to address these challenges.

AL, a subfield of artificial intelligence, encompasses an iterative feedback process that selects valuable data for labeling on the basis of model-generated assumptions and uses this labeled data to iteratively enhance the model's performance.^(p6) The fundamental focus of AL research revolves around creating a series of well-motivated functions based on model-generated hypotheses to guide data selection.^(p7) These selection functions can pinpoint the most valuable data in the database, facilitating the construction of high-quality ML models or the discovery of more desirable molecules with fewer labeled experiments. Additionally, they aid in selecting the most informative data from labeled datasets, eliminating redundancy and promoting the creation of a balanced training set. These advantages of AL neatly align with the challenges faced in drug discovery, making the integration of AL a valuable facilitator throughout the drug discovery process.

The concept of AL has been in existence for nearly four decades and was introduced into the field of drug discovery approximately two decades ago.^{(p8),(p9),(p10)} However, the early application of AL in drug discovery was relatively limited due to the incompatibility between its infrastructure and the rigidity of high-throughput screening.^{(p11),(p12)} In recent years, with advancements in automation technology for high-throughput screening and improvements in the accuracy of ML algorithms, the application of AL algorithms in the field of drug discovery has rapidly developed.^{(p13),(p14)} Despite the growing importance of AL in drug discovery, there is currently a lack of comprehensive reviews that serve as valuable references to provide insights and directions for experts seeking to effectively apply AL in the field of drug discovery. To bridge this gap, this article presents a comprehensive review that explores the application of AL across different stages of the drug discovery process and discusses the challenges and potential future directions.

In this article, we begin by providing a comprehensive introduction to the AL workflow. We then conduct a systematic review of the applications of AL in various facets of drug discovery as documented in recent decades. Finally, we summarize the current status and constraints of AL within drug discovery while also providing a prospective outlook on the development of AL in this domain.

The workflow of AL

AL is a dynamic feedback process that commences with the creation of a model using a limited set of labeled training data. It subsequently iteratively selects informative data points for labeling from the dataset based on model-generated hypotheses, employing a well-defined query strategy. The model is then updated by integrating these newly labeled data points into the training set during each iteration. The AL process culminates when it attains a suitable stopping point, ensuring an efficient and accurate model. As such, AL typically comprises four pivotal components: the initial training set, the ML algorithm, the query

strategy, and the evaluation indicators used to determine the stopping point.^(p15) The following section and [Figure 1](#) provide a comprehensive elaboration on each of these components, offering detailed insights into their roles and contributions within the AL framework.

The establishment of the initial training set serves as the cornerstone of the AL process, wielding a substantial influence over its course. The current landscape of extensive research literature highlights a prevailing trend: the selection of the initial training set is different between prospective and retrospective studies,^{(p16),(p17),(p18),(p19),(p20)} as detailed in [Figure 1](#). It is worth noting that the choice of the initial training set can significantly impact the behavior and outcomes of AL. However, it is essential to underscore that all AL projects, regardless of the timing of the implementation, have demonstrated significant advantages for those that embrace an adaptive experimental design. This highlights the pivotal role of AL in achieving more effective results, whether it is integrated early or at a later stage in the study.^(p11)

The selection of the ML algorithm is a critical aspect of an AL workflow. AL has successfully incorporated a range of popular ML algorithms, including traditional ML,^{(p8),(p16)} deep learning (DL),^{(p21),(p22),(p23),(p24)} and more sophisticated algorithms^{(p25),(p26)}; details are shown in [Figure 1](#). This underscores the flexibility and expanding possibilities in ML algorithm choices for AL. This integration of diverse ML algorithms has also widened the arsenal available to experts implementing AL and encouraged researchers to synergize AL with advanced and more accurate ML algorithms.^(p11) This fusion holds great promise for enhancing the efficacy of AL in drug discovery. Nevertheless, it is crucial to acknowledge that the performance of a specific ML algorithm can vary significantly across distinct research challenges. As a result, researchers must exercise diligence and meticulously select the most appropriate ML algorithm when tackling specific issues within the domain of drug discovery.

The heart of AL lies in the choice of query strategies used to identify valuable data. Typically, these strategies can be broadly categorized into three groups: exploitative, explorative, and balanced query strategies, shown in [Figure 1](#).^(p7) Exploitative query strategies prioritize the selection of data with potentially desirable traits, such as molecules with high activity potential against specific targets, without necessarily considering their potential to enhance the model. In contrast, explorative query strategies focus on choosing molecules that can provide fresh insights into the model, even if they may lack the sought-after attributes. These strategies aim to expand the model's understanding and knowledge. To strike a balance between exploitation and exploration and select data with desirable properties while enhancing the model's performance, various balanced selection strategies have emerged. For example, the simplest balanced approach involves choosing an equal mix of exploitative and explorative data.^(p27) Moreover, more sophisticated methods have been developed, incorporating weight coefficients to fine-tune the equilibrium between exploration and exploitation.^(p16) Considering the diverse effects of these query strategies, it is imperative for researchers to carefully select the strategy that aligns with their precise research objectives and goals. The choice of the query strategy should be driven by the specific aims of the AL project to maximize its effectiveness.

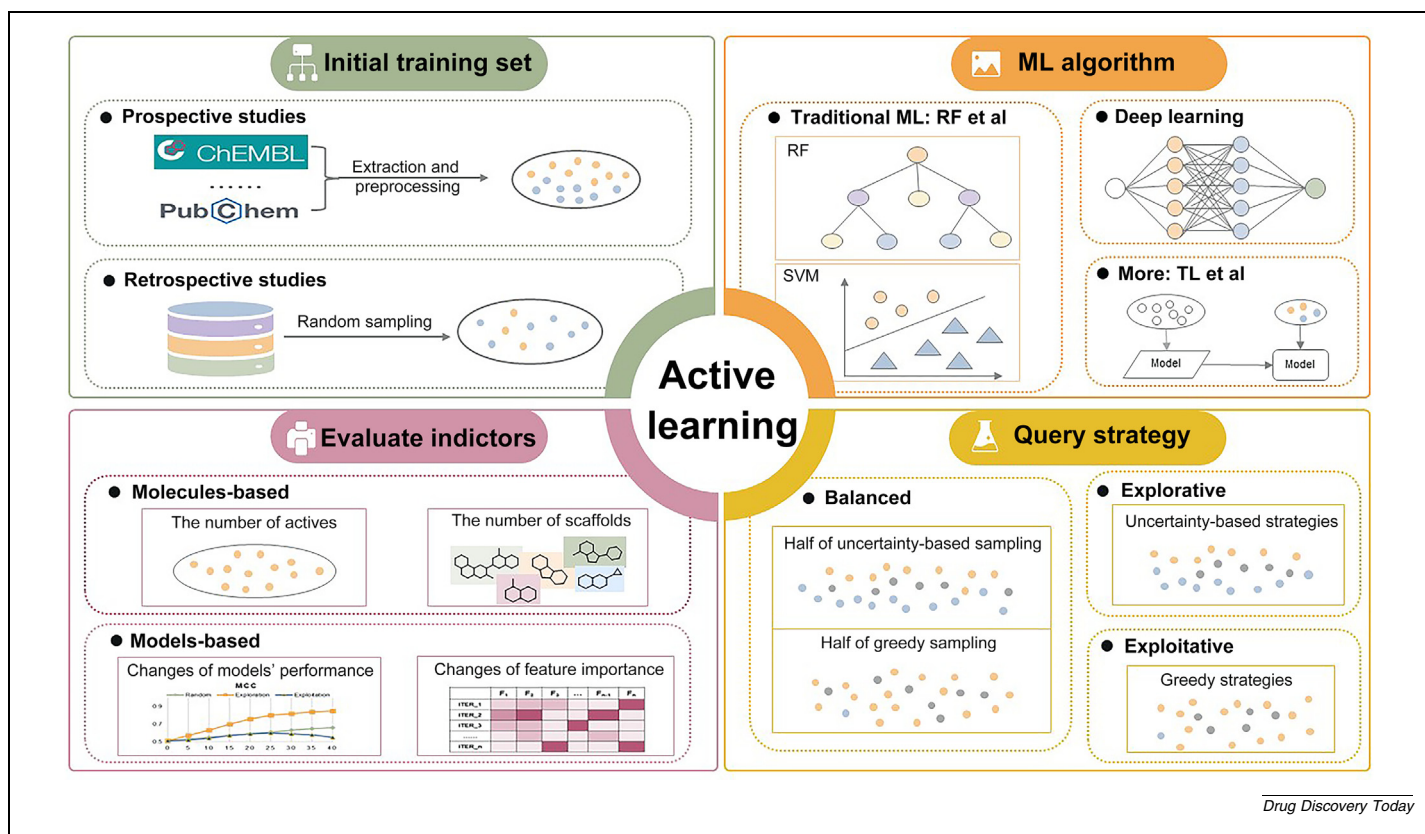


FIGURE 1

The overview of active learning. AL involves an iterative feedback process comprising four main components. The first is initial training set selection, differing between prospective and retrospective studies. Prospective studies usually begin with preprocessed historical data, whereas retrospective studies typically start with randomly selected small datasets. The second component involves ML algorithms. Various algorithms have been applied successfully to AL workflow, encompassing traditional ML algorithms, such as random forest (RF) and support vector machine (SVM), advanced DL algorithms such as deep neural networks (DNNs), and more refined algorithms such as transfer learning (TL). The third part comprises query strategies guiding data selection, categorized into three types: explorative strategies, such as uncertainty-based strategies; exploitative strategies, such as greedy strategy; and balanced strategies, such as half of uncertainty-based sampling and half of greedy sampling. The final part involves the evaluation metrics to determine whether to halt the AL learning cycle. These metrics are divided into molecule-based evaluation metrics, which primarily revolve around the queried data, such as the number of actives and active scaffolds retrieved, and model-based evaluation metrics, which primarily focus on changes within the model itself, such as changes in model performance and the importance of features.

The concluding phase of an AL campaign involves the critical decision of when to terminate the workflow. This decision is closely linked to the evaluation indicators used to gauge the benefits of the AL process. Typically, these evaluation indicators can be categorized into two main groups: molecule-based indicators^{(p20),(p28)} and model-based indicators.^{(p16),(p29)} Details are shown in Figure 1. These indicators shed light on how the model is evolving during the AL process. However, it is important to note that these indicators primarily assess the current iteration's status and do not directly measure the potential gains that could be achieved from additional iterations. To address this limitation, some researchers have recently turned to analytical modeling and statistics to estimate the potential benefits of further AL rounds.^{(p19),(p30)} These estimates serve as additional indicators to guide the decision on whether to continue iterating or conclude the workflow. Ultimately, researchers can determine when to halt the AL workflow based on these indicators, aligning their decision with their specific needs and objectives.

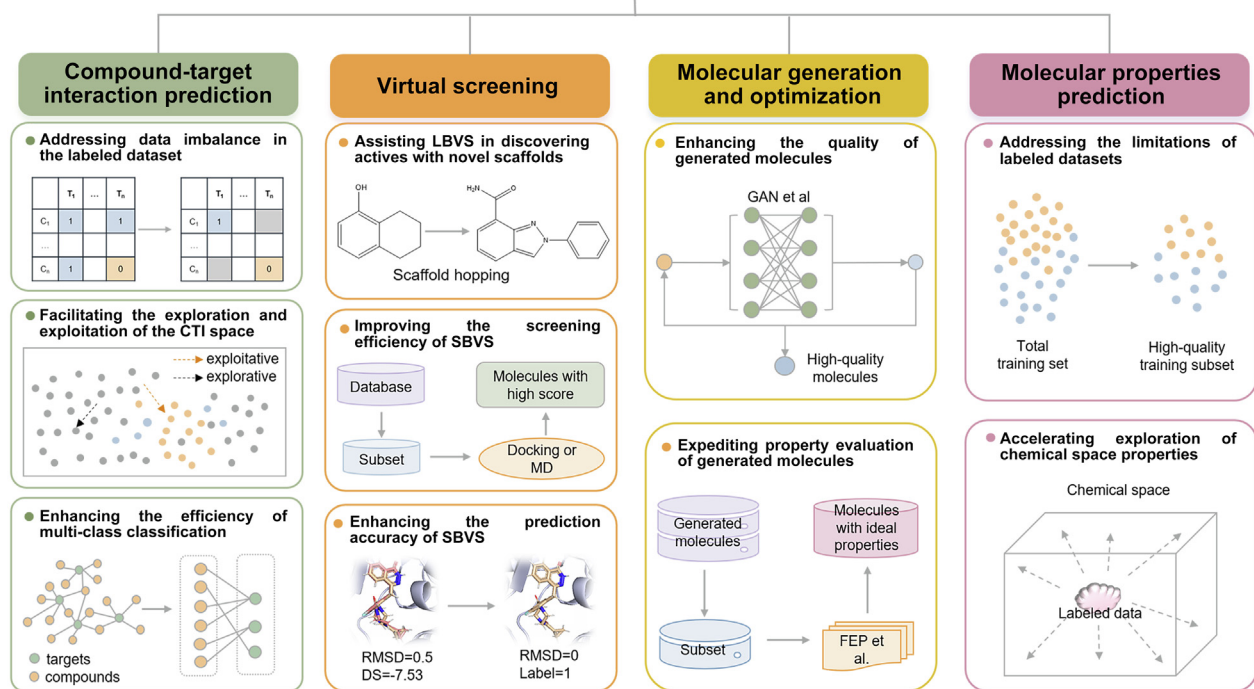
Application of AL

AL, with its integration of diverse ML algorithms and query strategies, has demonstrated remarkable effectiveness across various critical stages of drug discovery. These stages encompass the prediction of compound–target interactions (CTIs), virtual screening (VS), molecular generation and optimization, and molecular property prediction, among others. These applications serve distinct purposes and collectively contribute to the advancement of the drug discovery process. Specific details are provided in Figure 2. In this section, we offer a comprehensive and systematic review of the research conducted on the application of AL in different stages of drug discovery.

CTI prediction

Accurate identification of CTIs is crucial for various applications, including drug repurposing, target fishing, and predicting adverse drug reactions.^{(p31),(p32)} Nevertheless, the sheer magnitude of the CTI search space renders experimental measurements

The application of active learning in the field of drug discovery



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FIGURE 2

The main application of AL in various stages of drug discovery. In CTI prediction, AL has been used successfully to address data imbalance in the labeled dataset, facilitate the exploration and exploitation of the CTI space, and enhance the efficiency of multiclass classification. In VS, AL has bridged the major shortcomings of the two VS methods: It assists LBVS in discovering actives with novel scaffolds, improves the screening efficiency of SBVS, and enhances the prediction accuracy of SBVS. In the phase of molecule generation and optimization, AL has been used to enhance the quality of generated molecules and expedite property evaluation of the generated molecules. In molecular property prediction, AL addresses the limitations of labeled datasets and accelerates the exploration of the chemical space properties.

impracticable. ML has sped up the evaluation of CTIs, but its predictive precision is hindered by the imbalanced distribution of data within labeled datasets. Furthermore, given the expansive interaction space, relying solely on existing labeled data to construct the model is insufficient for accurately predicting all CTIs. Additionally, devising efficient solutions for complex multiclass classification challenges linked with CTIs while minimizing the need for expert intervention represents an ongoing hurdle in CTI research. The incorporation of AL with diverse query strategies has shown promise in mitigating these challenges (see Figure 3), as evidenced by numerous successful applications.

Addressing data imbalance in the labeled dataset

One of the primary factors contributing to inaccuracies in predicting CTIs is the uneven distribution of labeled data. This includes imbalances between interacting and noninteracting pairs, as well as variations in the distribution of interaction data across different targets. Addressing these issues and constructing training sets with balanced interaction and noninteraction ratios, along with extensive target coverage, are pivotal for the development of high-quality interaction prediction models. AL with an explorative selection strategy provides an effective solution by iteratively selecting valuable data to enrich the model with new information, a method that has been well documented

in relevant literature as an efficient means of addressing data imbalance challenges.^{(p19),(p33),(p34)}

A pioneering piece of research conducted by Reker *et al.*^(p19) introduced an AL framework to extract a balanced training set from an imbalanced dataset and evaluated its effectiveness on three datasets with imbalanced CTIs. The AL workflow iteratively selects molecule-target pairs from the dataset using various query strategies such as curious, greedy, or random to expand the training set and update the RF classification model. Test results revealed that the curious query strategy, which prioritizes the selection of the most uncertain molecule-target pairs in each iteration, significantly contributed to the creation of a well-balanced training set with broad target coverage. In another study, this method also demonstrated its effectiveness in scenarios with sparse labeled data, underscoring AL's ability to construct a well-balanced training set.^(p34) Furthermore, the Sharma *et al.*^(p33) study also showcased the feasibility of mitigating class imbalance in CTI prediction through the implementation of an AL strategy. This approach involves iteratively selecting points closest to the boundary to update the training set and refine the model. As a result, it successfully addresses the issue of class imbalance, culminating in the development of a high-quality CTI prediction model.

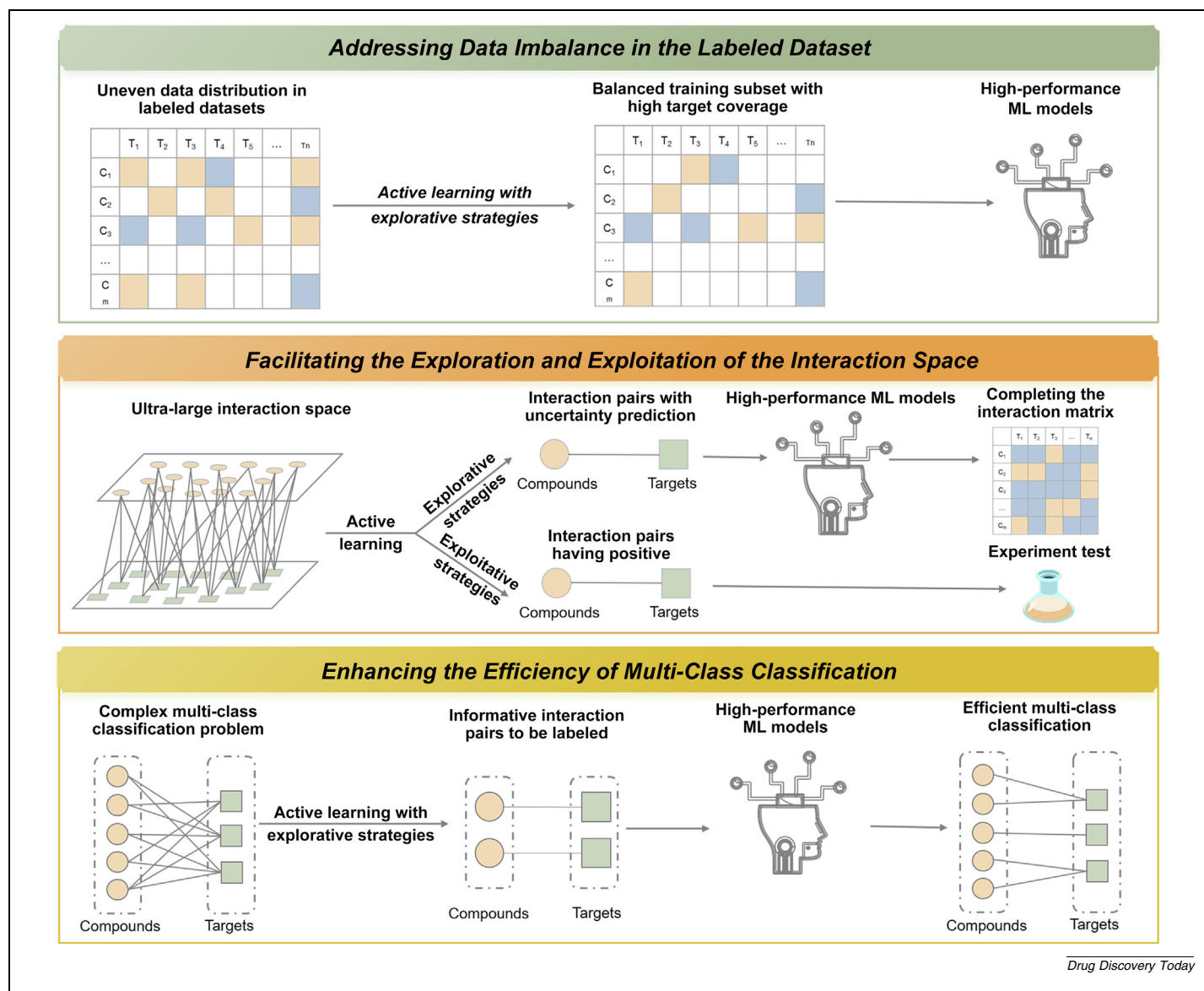


FIGURE 3

Schematic representation of AL resolves challenges in CTI prediction. First, AL employing an explorative selection strategy addresses data imbalance in the labeled datasets by iteratively selecting interactive pairs with inaccurate predictions to add to the training set, thus extracting a balanced training set. Second, AL facilitates the exploration and exploitation of the CTI space. The explorative selection strategy iteratively selects informative interaction pairs to efficiently explore the interaction space, constructing high-quality interaction prediction models. Conversely, the exploitative selection strategy iteratively selects interactive CTI pairs, efficiently exploiting the interaction space. Third, AL employing an explorative selection strategy iteratively selects the information-rich data to construct high-quality multiclass classification models, thereby accurately classifying multiple molecules into multiple targets or multiple binding sites, efficiently solving the multiclass classification problems related to CTI.

These findings highlight that AL with an explorative selection strategy efficiently mitigates data distribution imbalances to assemble a well-balanced training set, encompassing balanced interacting and noninteracting pairs and comprehensive target coverage, regardless of whether the labeled dataset is data-rich or data-scarce. However, it is worth noting that the Reker *et al.*^(p19) study indicated that although the curious selection strategy achieved broad target coverage, it may not be as rapid as random sampling in terms of target coverage speed. Therefore, there is room for further research and the development of new strategies to enhance efficiency in this regard.

Facilitating the exploration and exploitation of the CTI space

ML-based prediction of CTIs allows the rapid assessment of CTIs. However, the accuracy of these predictions significantly depends on the quality of the training set. To construct a model capable of accurately predicting all interactions within the vast interaction space, it is crucial to identify a training set that comprehensively captures the properties of this space. Randomly labeling molecule-target pairs for training set construction can lead to significant resource wastage. Therefore, the integration of an algorithm that efficiently guides labeling experiments, facilitating the exploration of the interaction space and the swift generation

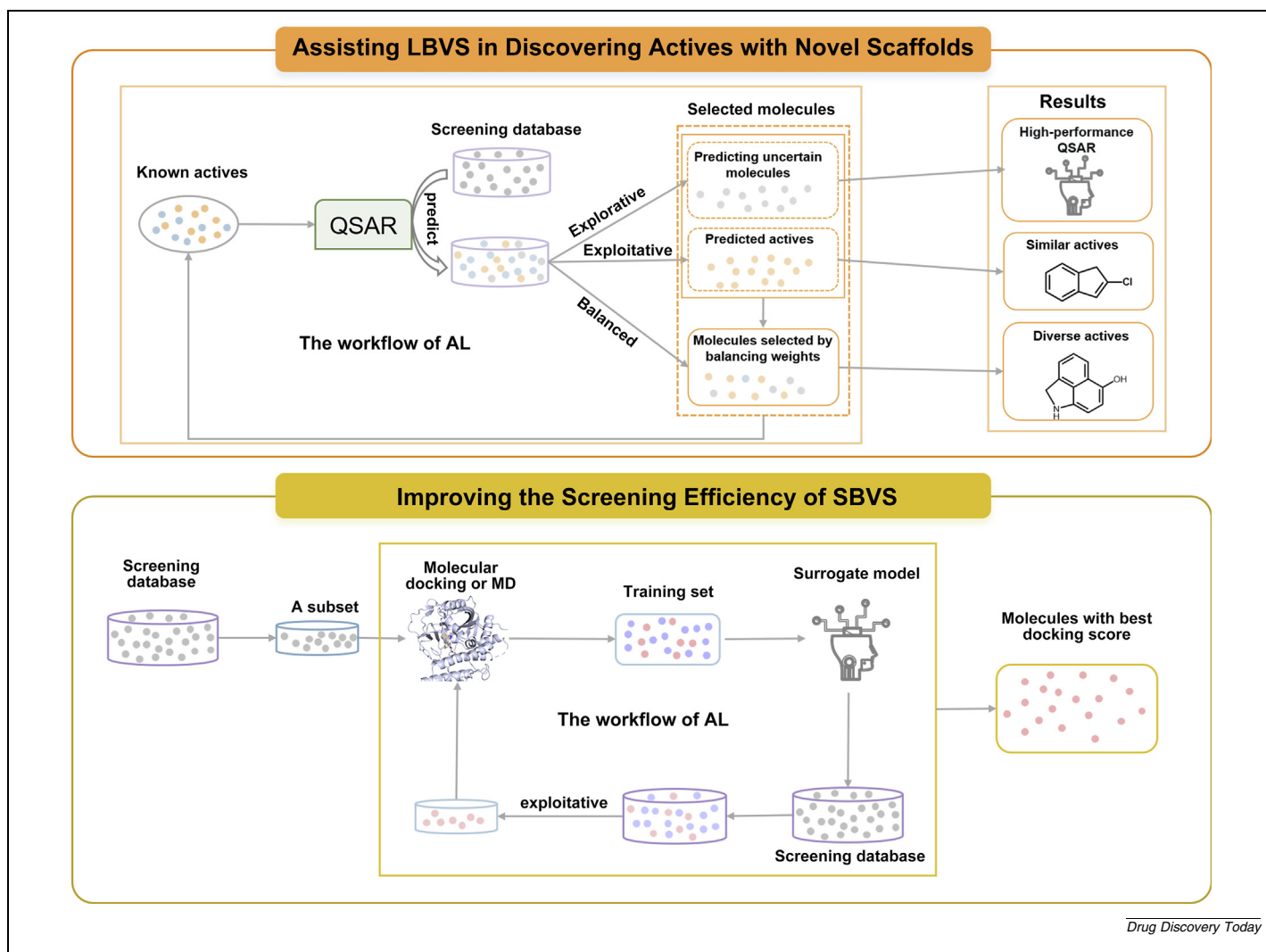


FIGURE 4

Schematic representation of AL mitigates the main drawback of two types of VS method. For LBVS, AL aids in discovering actives with novel scaffolds. The explorative strategies iteratively select molecules with uncertain predictions to add new information to the model. The exploitative strategy iteratively selects molecules most likely to be active, increasing the discovery of actives. The balanced selection strategy prioritizes molecules with both new information and desirable properties, facilitating the discovery of actives with novel scaffolds. For SBVS, AL employing the exploitative selection strategy improves screening efficiency by iteratively selecting molecules likely to have desired scores for computation, enabling the quick identification of high-scoring molecules from the database with minimal computational resources.

of a representative training set, is essential. AL based on the explorative query strategy serves as a valuable algorithm for achieving this objective, with numerous reports highlighting its successful application.^{(p35),(p36),(p37),(p38),(p39)}

Two studies conducted by Naik *et al.*^{(p35),(p36)} effectively showcase the impact of AL. In their first study, they introduced an AL workflow that integrates an explorative query strategy with structure learning.^(p35) This workflow iteratively selected interaction pairs with low predictive confidence for annotation, expanding the training set and updating the structure learning model for predicting CTI phenotypes. The retrospective study revealed that AL based on the explorative query strategy can effectively guide experimental decisions, resulting in a substantial reduction in the number of labeled experiments required to construct a model capable of accurately predicting the overall dataset. Building on the first study, Naik *et al.*^(p36) designed an active learner equipped

with liquid handling robotics and automated microscopy to prospectively investigate the impact of 48 compounds on the subcellular localization of 48 proteins. This active learner predicts condition-target phenotypes based on the principle that similar conditions exhibit the same phenotypes with the same type of target. During each iteration, condition-target combinations not predicted by the model are prioritized for experimental validation, facilitating iterative model updates. The final outcomes demonstrated that this active learner efficiently constructs a model that accurately captures all phenotypes, requiring only 29% of all possible experiments. Furthermore, Sun *et al.*^(p38) iteratively selected informative data through a hybrid explorative query strategy, which selected half of the data on the basis of least predictive confidence score and the other half on the basis of highest entropy, to swiftly improve the performance of their proposed CTI prediction model. The results of retrospective

study demonstrated that this approach enables the construction of high-performance interaction prediction models with significantly less training data. Additionally, their study also showed that the iterative selection of informative data through specific explorative query strategies can swiftly enhance the performance of some other interaction prediction models, allowing the development of high-quality prediction models with a substantially reduced need for labeled data.

Although guided space exploration to select informative molecules for constructing high-quality ML models is important, the primary objective of many studies is the discovery of compound-target pairs with interactions. The introduction of AL based on the exploitative query strategy, which iteratively selects the most promising compound-target pairs based on model-generated hypotheses, can rapidly identify a substantial number of interacting molecule-target pairs within the interaction space and can efficiently guide the exploitation of this space, as demonstrated in the study by Kanga *et al.*^(p27) In their research, they iteratively selected a batch of molecular target pairs using a specific query strategy for labeling, expanding the training set and reconstructing the interaction prediction model. Retrospective validation on a subset extracted from PubChem revealed that the greedy strategy significantly outperformed random and other query strategies in selecting more interacting pairs. However, they noted that the greedy strategy had a smaller impact on model performance than the uncertainty strategy. To address this, they proposed a balanced query strategy, selecting half of the data on the basis of the greedy strategy and half on the basis of the uncertainty strategy. Results showed that this query strategy can improve the model's performance and retrieve more pairs with interactions simultaneously. In addition, the study by Luo *et al.*^(p40) also demonstrated that using the UCB query function, a balanced query strategy, to iteratively select interaction pairs with high binding affinity and high uncertainty for labeling can enhance the model's performance and recall more strong-binding compound-target pairs simultaneously.

These studies have demonstrated that a balanced query strategy, which integrates both explorative and exploitative query strategies, can effectively recover a greater number of molecule-target pairs with interactions while enhancing the model's performance.^{(p27),(p40)} This suggests its ability to guide both the exploitation and exploration of the interaction space simultaneously. However, in the field of predicting CTIs, there is a relative dearth of research in this area, and the presently used balanced selection strategies are rather rudimentary. Hence, there exists a need to introduce or develop novel balanced selection strategies in this domain to better facilitate the guidance of both exploitation and exploration within the interaction space.

Enhancing the efficiency of multiclass classification

In the field of interaction studies, complex scenarios often arise where multiple compounds exhibit unknown interactions with various targets or binding sites, posing a formidable multiclass classification challenge in accurately assigning drugs to specific targets or binding sites. ML has proved to be effective in addressing this challenge by leveraging known-type training data to construct classification models that swiftly categorize data into relevant classes. However, the construction of the training data through experiments or expert labeling of the data categories

consumes considerable time and resources. Therefore, the research focus is on constructing high-performance classification models with minimal training data. AL based on the explorative selection strategy accomplishes this objective by iteratively selecting the most valuable data to improve the classification model's performance, as substantiated by relevant research.^{(p41),(p42)}

Lang *et al.*^(p41) introduced the first AL framework to tackle multiclass classification challenges within the domain of CTIs. They approached the multiclass classification issue as a series of binary classification tasks (one versus all), employing SVM to compute the likelihood of molecules falling into specific categories. Within the AL workflow, they adopted the uncertainty selection strategy as the query strategy to select molecules with the smallest discrepancy between the two highest positive probabilities for labeling, then using these labeled points to iteratively refine the model. This AL workflow was tested on 15 multiclass classification scenarios, demonstrating the ability to construct high-quality multiclass classification models with significantly fewer training samples through AL. In essence, AL efficiently reduced the need for labeled training data in solving complex multiclass classification problems in the realm of CTIs.

Furthermore, a study by Rodríguez-Pérez *et al.*^(p42) demonstrated the efficiency of AL workflows in accurately determining the binding sites for a set of molecules across multiple sites with limited sample annotations. They employed an entropy-based molecule selection strategy in each AL iteration to identify informative data, thereby enhancing a multiclass RF classification model. This approach resulted in the development of a high-quality multiclass classification model with reduced samples, facilitating the precise classification of kinase inhibitors to their respective binding sites. This demonstrates the ability of AL to improve the efficiency of solving multiclass classification problems of assigning molecules to the correct active sites. Additionally, they leveraged the model's performance improvement during AL iterations to evaluate the information content of different representations, introducing a novel application of AL.

VS

VS can swiftly identify potential actives within the vast space to significantly mitigate the overall costs associated with drug development. Methodologically, VS falls into two main approaches: ligand-based virtual screening (LBVS) and structure-based virtual screening (SBVS). LBVS strategies, such as structure-activity relationship (SAR), assume structurally similar molecules share analogous biological properties, limiting the identification of hits with novel scaffolds. In contrast, SBVS employs computational simulations to mimic ligand-protein binding interactions, offering an advantage in identifying actives with unseen scaffolds. However, SBVS methods, especially molecular dynamics (MD) simulations, can be resource-intensive. Additionally, widely used SBVS method molecular docking suffers from low predictive accuracy, posing challenges in drug discovery. To address the shortcomings inherent in both LBVS and SBVS, an emerging solution known as AL has garnered significant attention, as depicted in Figure 4. Numerous studies have reported the efficacy of AL in enhancing the efficiency and accuracy of VS.

Assisting LBVS in discovering actives with novel scaffolds

LBVS methods, guided by the similarity principle, identify hits that share structural similarities with known ligands from databases. This approach limits the discovery of actives with novel scaffolds. However, identifying novel actives is essential in drug screening, especially when considering patent protection issues. Therefore, the focus of LBVS research is on devising strategies to overcome the limitations and identify hits with innovative scaffolds. Among LBVS methods, SAR modeling is frequently employed, with the screening ability of SAR models closely tied to their performance. Efforts have been concentrated on guiding SAR models to discover more potential actives with novel scaffolds. To this end, the integration of AL based on balanced selection strategies has proved to be a valuable approach, as demonstrated in a reported study.^(p16)

As early as 2003, Warmuth *et al.*^(p8) showed that AL based on an exploitative query strategy significantly aids SAR models in retrieving more actives. They conducted a retrospective study using thrombin data, and the results demonstrated that the strategy favoring molecules with the highest positive score at each iteration exhibited superior performance in retrieving more actives. However, the novelty and diversity of the retrieved actives were not addressed in this study. Numerous studies have also highlighted the effectiveness of AL based on explorative selection strategies in improving SAR model performance by introducing novel information.^{(p20),(p43),(p44)} For example, Czarnecki *et al.*^(p43) introduced an explorative query strategy known as the CSJ sampling strategy, which combines clustering techniques and quasi-greedy approaches^(p45) to select a diverse group of molecules that contribute new information to the model. Fujiwara *et al.*^(p20) iteratively selected prediction inaccurate molecules through an explorative query strategy similar to the 'query by committee' (QBC) strategy to rapidly improve the performance of the model.

Researchers have also explored the idea of a balanced query strategy that combines exploitative and explorative selection strategies to enhance SAR model performance and retrieve actives with novel scaffolds.^(p16) A study by Reker *et al.*^(p16) provides substantial evidence that an AL workflow based on a balanced query strategy can significantly enhance SAR models in rapidly identifying actives with novel structures. This balanced strategy considers multiple factors, such as predicted activity, RF similarity with training data, and model prediction uncertainty. The research by Reker and colleagues demonstrated that this balanced selection strategy consistently outperformed other strategies in discovering structurally novel actives. Furthermore, it was effectively employed for prospective screening of CXCR4 inhibitors, emphasizing its capability to enrich models with new information and identify compounds with innovative scaffolds. However, it is important to note that the exploration of AL for scaffold hopping in LBVS remains relatively uncharted territory with limited relevant studies. Additional research and the development of balance-based selection strategies are needed to effectively support the discovery of structurally novel actives using LBVS. Furthermore, promoting the synergistic application of AL in conjunction with LBVS is a priority for further exploration and development in this field.

Improving the screening efficiency of SBVS

SBVS methods are highly regarded for their ability to efficiently identify compounds with novel scaffolds. However, they often grapple with slow screening speeds. Even relatively fast molecular docking processes can be resource-intensive when applied to ultra-large databases.^(p46) This bottleneck arises from the significant computational resources expended on evaluating poorly scored molecules during the screening process. As a result, research efforts are concentrated on improving the efficiency of SBVS by developing strategies that intelligently guide the exploitation of chemical space, directing computational resources toward the most promising subset of molecules. AL with exploitative or balanced query strategies has proved to be effective in achieving this goal, and numerous studies have validated its effectiveness.

Early on, researchers introduced AL to enhance the efficiency of molecular docking screening.^{(p21),(p47),(p48),(p49),(p50)} For instance, Graff *et al.*^(p21) introduced the Bayesian optimization algorithm, an AL strategy that iteratively selects data for docking on the basis of assumptions generated by a surrogate model. This approach mitigates the increasing computational cost of screening growing compound databases, thus improving the efficiency of the docking screening process. Their study systematically examined various surrogate models, acquisition functions, and batch sizes. The results showed that Bayesian optimization effectively identifies top-ranking docking score molecules, focusing computations on the most promising segment of the database and accelerating the screening process. Notably, the greedy query, a purely exploitative query strategy, performed the best in recalling the top-scoring molecules. In addition, the study by Yang *et al.*^(p48) demonstrated that AL based on a novel balanced query strategy, which focuses on selecting the top 0.1% of molecules with the highest uncertainty from the top 5% of compounds based on ML scores, can improve the performance of the docking score prediction model and effectively recall most molecules with the highest docking scores.

Compared with molecular docking, MD simulations offer higher accuracy in predicting binding affinity but require significantly more computational resources, limiting their applicability to a select subset of molecules. Therefore, it is crucial to prioritize MD calculations for molecules with the highest potential. Building on the success of combining AL with molecular docking, researchers have extended this approach to integrate AL with MD simulations. This strategic alignment aims to focus MD calculations on the most promising molecules, efficiently identifying a majority of high binding affinity molecules by simulating only a small fraction of the molecules. This approach has received support from numerous reports.^{(p18),(p51),(p52),(p53),(p54)}

A notable example is the study conducted by Gusev *et al.*,^(p18) who employed an AutoML-AL workflow to conduct MD screening on a library of 8000 molecules sharing the same scaffold as the ligands of the SARS-CoV-2 papain-like protease. This workflow first uses a balanced selection strategy to choose structurally diverse molecules with favorable prediction scores, with the goal of identifying promising molecules while exploring the chemical space. It then transitions to an exploitative strategy for selecting the most promising molecules. The results demonstrated that the

balanced and exploitative selection strategies can find more molecules with better relative binding free energy (RBFE) than random selection, and this workflow efficiently screens a large number of molecules with higher binding affinities than the reference molecule by simulating only a few hundred molecules.

Furthermore, a retrospective study conducted by Thompson *et al.*^(p52) on a database comprising 1000 molecules sharing the same backbone and featuring well-defined RBFE values for TYK2 showcased the remarkable capability of AL. The study demonstrated that AL can efficiently recall a substantial portion of molecules ranked within the top 100 in terms of RBFE by computing only a small number of molecules, regardless of the specific algorithm and molecule selection strategy employed.

The aforementioned studies provide a comprehensive illustration of the effectiveness of AL in guiding space exploitation, swiftly identifying the most promising molecules for subsequent computational prediction, thereby enhancing the efficiency of SBVS. Capitalizing on this advantage of AL, Schrödinger has developed various software applications, including AL-Glide and AL-FEP+, designed to accelerate SBVS by incorporating AL workflows. These applications automate the screening process, enhancing accessibility and user-friendliness for broader application.

Enhancing the prediction accuracy of SBVS

In addition to focusing on the screening speed of SBVS, research efforts are predominantly centered on enhancing screening accuracy and reducing the incidence of false positives. Molecular docking, as the most widely used SBVS method, has been identified in numerous studies as having a high false-positive rate.^(p55) Therefore, it is of paramount importance to improve its screening accuracy. The accuracy of molecular docking primarily hinges on two key factors: the ability to predict the protein–ligand binding conformation accurately and the precision of binding affinity scoring.^(p56) AL has now been introduced to bolster the accuracy of molecular docking from these two critical aspects, as evidenced by various studies.^{(p57),(p58),(p59),(p60),(p61)}

In the quest to enhance conformational search, Hsu *et al.*^(p59) implemented AL to augment the predictive performance and precision of protein–ligand binding pose predictions using the tinyIFD method. This approach involves constructing classification models for pose prediction for each protein in the training set, culminating in the creation of a model ensemble. When predicting the binding pose for a novel protein and compound, a specific number of classification models are selected on the basis of their similarity to proteins and ligands in the training set, thus enabling the prediction of the binding pose. The predictions of each model are then integrated to generate the final prediction results. The introduction of explorative-based AL extends the training set to include new protein families, ultimately enhancing the model's accuracy in predicting binding poses for novel protein and compound pairs.

Regarding the enhancement of the predictive accuracy of the scoring function, Wang *et al.*^(p61) introduced an AL approach based on a negative molecule selection strategy. This, when combined with the machine learning-based scoring function (MLSF), iteratively reduces the differentiation between inactive and active molecules within the training set. Consequently, this iterative process refines the MLSF, resulting in more accurate predic-

tions of a molecule's activity and a concurrent reduction in the false-positive rate during screening. By addressing these two fundamental aspects of molecular docking accuracy through AL, researchers are taking significant strides toward improving the overall efficiency and reliability of docking.

Additionally, de Oliveira *et al.*^(p62) also introduced an AL workflow to search the parameter space of the free energy perturbation (FEP) protocols, aiming to develop accurate FEP protocols for targets whose binding affinity cannot be precisely calculated under the default parameters. This AL workflow iteratively selects combination of parameters predicted to produce the smallest computational errors, ultimately identifying the most suitable parameters of FEP procedure for the particular target, thereby enabling more accurate calculation of the binding affinity. This study demonstrated that AL can improve the accuracy of FEP calculation for some special target and also proposed a new application aspect of AL.

Molecular generation and optimization

With the rapid advancements in DL algorithms, the generation and optimization of molecules have garnered significant attention from drug discovery experts. Typically, the process of generating and optimizing molecules primarily involves two key steps. First, it entails generating molecules with desired properties using diverse generative models such as recurrent neural networks and generative adversarial networks.^{(p63),(p64)} Second, it involves evaluating these generated molecules using experimental or precise computational methods to identify those that possess the desired properties. Consequently, the two primary research directions in molecular generation and optimization revolve around how to generate high-quality molecules and how to expedite the property evaluation of these generated molecules. AL has proved successful in both directions, enhancing the quality of generated molecules and accelerating the property evaluation of the generated molecules.

Enhancing the quality of generated molecules

Traditionally, simple molecular generative models used to create molecules with desired properties require substantial training data. However, the reality is that such training data are often limited. Researchers have integrated reinforcement learning (RL) and TL techniques to address this challenge and have achieved some success.^{(p65),(p66),(p67)} However, the effectiveness of TL and RL is also closely linked to the quality of the training data. Therefore, constructing a high-quality training dataset is crucial for generating molecules with desirable properties. AL has demonstrated the potential to facilitate the generation of molecules with improved properties, as indicated in the literature.^{(p25),(p26),(p68)}

For instance, Iovanac *et al.*^(p68) illustrated that a single generative model can be guided to produce more molecules with desirable properties through AL. In their study, they employed the Grammar Variational Autoencoder^(p69) as their generative model. During each iteration of the AL process, they conducted preliminary calculations to select structurally valid and novel molecules that met multiple property requirements from a pool of 100 000 molecules generated by the model. These selected molecules were combined with a random subset drawn from the training set from the previous iteration, maintaining a consistent total

of 100 000 molecules in the training set. This updated training set was then used to retrain the generative model. The final results clearly demonstrated that, after several iterations of AL, the generative model significantly improved its ability to generate more molecules with multiple desirable properties simultaneously.

Furthermore, Bengio *et al.*^(p26) showcased the effectiveness of AL in enhancing the generation of diverse molecules with desirable properties within RL models. Their study introduced GFlow-Net, a novel RL algorithm capable of sampling trajectories based on a specified reward function, thus promoting space exploration and generating a more diverse range of target molecules. During each iteration of their AL workflow, they employed Auto-Dock to assess the binding affinities of model-generated molecules for the target. This binding affinity data was then used to iteratively update the generative model. The results from multiple iterations illustrated the model's responsiveness to AL guidance in producing more diverse molecules with higher binding affinities. Remarkably, the authors also used the quality of the molecules generated by the model after each iteration to highlight GFlowNet's superior performance in generating diversity.

Expediting property evaluation

Accurately assessing the properties and activities of generated molecules to identify those with desirable attributes is of utmost importance. Traditional methods, such as experimental testing or high-precision calculations, are known to be time-consuming and expensive. Therefore, it is essential to develop efficient approaches for property assessment that allow the rapid identification of promising molecules. In the context of VS, previous research has shown that AL can enhance screening efficiency, facilitating the swift identification of the most promising molecules. Researchers have responded by incorporating AL, specifically using the exploitative query strategy, to expedite the property evaluation of model-generated molecules and to identify molecules with desirable properties while conserving computational resources.^{(p70),(p71),(p72),(p73),(p74)}

For example, Konze *et al.*^(p70) introduced AL to accelerate the assessment of binding affinities between the generated molecules and their target receptors. They initiated their AL workflow by randomly selecting 1000 molecules from the generated dataset for FEP calculations, establishing the initial training set for constructing an alternative model. In each subsequent iteration, they employed a greedy query strategy to select the most promising molecules for FEP calculations. These selected molecules were then used to expand the training set and iteratively update the alternative model. This approach successfully achieved the goal of swiftly identifying molecules with high binding affinities in the generated dataset while conserving computational resources.

In addition, some automated flow platforms have integrated AL to expedite the evaluation of the properties of generated molecules. For instance, Desai *et al.*^(p74) reported a flow technology platform that combines AL, microfluidic synthesis, and bioassay assays. In this AL workflow, RF regression models were constructed using initially collected actives as the training set. The process involved three AL stages to iteratively select valuable data and update the model. In the first stage, the 'most active undersampled' query strategy was used to select compounds with the lowest cumulative previous use, emphasizing the max-

imization of design space exploration and the rapid identification of regions with higher activity. The second phase employed the 'chase potency' query strategy to select molecules with the highest scores, aiming to further map areas of high activity. Finally, the 'combined' strategy was used to identify and optimize additional potential activity 'hot spots.' The ultimate outcome demonstrated that guided experimental selection through AL led to the discovery of a new AbI kinase inhibitor with high binding affinity, achieved by synthesizing and testing only 21 molecules. These findings underscore the potential of AL to identify molecules with high binding affinity while minimizing the number of experiments required.

Molecular properties prediction

In drug discovery, target molecules are expected to possess not only a strong binding affinity but also favorable druglike properties. Evaluating molecule properties is crucial, and ML plays a vital role in this assessment. However, the accuracy of ML predictions is often hindered by issues in labeled data, such as data redundancy. Furthermore, with the continuous expansion of the chemical space, relying solely on existing labeled datasets falls short of accurately predicting the properties of all molecules. Methodologies are needed to alleviate the limitations associated with labeled data and guide the selection of informative data from the vast chemical space, facilitating the construction of a high-quality training set for more accurate property predictions. AL with an explorative query strategy emerges as a potential solution to these challenges, and numerous reports have demonstrated its effectiveness.

Addressing the limitations of labeled datasets

The accuracy of predicting molecular properties hinges on the quality of the training set employed for constructing the prediction model. This training set typically comprises directly collected labeled data, but it often harbors redundant information that can negatively impact the model's performance. Therefore, the removal of such undesirable data is essential to create a high-quality training set and enhance the model performance. AL with an explorative selection strategy proves to be a valuable tool for the elimination of this redundancy. By consistently selecting valuable data from the dataset, this approach effectively prunes out redundancies, thereby creating high-quality training sets. A vivid illustration of this can be found in the Ding *et al.*^(p17) retrospective study on the plasma exposure dataset of orally administered drugs. The AL process proposed by them was initiated by randomly selecting an initial training set to construct a deep neural network classification model. Subsequently, it iteratively enhanced the training set by identifying uncertain predicted molecules through four uncertainty-based selection strategies, which included entropy sampling,^(p75) MC-dropout sampling,^(p76) feature space distance sampling,^(p77) and latent space distance sampling.^(p78) The final results underscore the remarkable capacity of AL based on explorative query strategies to guide the selection of valuable data and eliminate redundancy, ultimately resulting in the creation of high-quality models. Notably, entropy-based sampling proved particularly effective, achieving superior model performance with less than one-third of the data. This serves as compelling evidence that

AL can effectively eliminate redundant data from the training set.

Accelerating exploration of chemical space properties

ML plays a pivotal role in molecular property assessment, but its effectiveness hinges on the training data. When the model generates predictions that significantly deviate from the training data, accuracy is compromised. With the chemical space continuously expanding, relying solely on limited labeled data is insufficient to make accurate property predictions for all molecules. The incorporation of AL employing the explorative query strategy proves to be a potent approach. This method systematically explores the chemical space by iteratively identifying molecules that either receive uncertain predictions from the current model or significantly differ from those in the existing training set. This iterative process allows the addition of new information to the training set, predicting molecular properties more accurately. The efficacy of this approach has been corroborated through numerous studies.^{(p22),(p29),(p79),(p80),(p81)}

For instance, Gubaev *et al.*^(p79) introduced AL to construct high-quality property prediction models with less data to accurately predict the properties of molecules and reduce the prediction error for outliers. In their proposed AL workflow, they constructed a moment tensor model (MTM) by randomly selecting molecules as an initial training set to predict molecular properties based on the local atomic environment. At each iteration, a selection strategy that integrates the D-optimality criterion^(p82) was employed to select novel molecules in comparison with those in the training set, thereby expanding the training set and updating the models. The results of their study on the QM9 dataset demonstrated that, through AL, it is possible to construct models that accurately predict the properties of outliers with less experimental annotation. Zhang *et al.*^(p22) introduced a method for evaluating prediction uncertainty through a Bayesian semisupervised graphical neural network. Their study demonstrated that iteratively selecting molecules inaccurately predicted by the current model through AL can effectively enhance the model's performance and improve accuracy in predicting outliers.

Additionally, Hao *et al.*^(p80) introduced the Active Semi-supervised Graph Neural Network (ASGN) framework, which employs AL to iteratively identify informative data and construct high-performance property prediction models with fewer labeled experiments. ASGN uses a teacher-student framework, where the teacher model learns general molecular representations from all available data and the student model fine-tunes its parameters using labeled data to predict the properties of unlabeled data. At each iteration, a diversity-based molecule selection strategy is applied to choose new representative data, thus expanding the training set and updating the model. The performance of the ASGN framework was assessed on the QM9 and OPV datasets, demonstrating that AL can guide data selection to predict the properties of molecules more accurately with fewer experimental labeling requirements. The aforementioned studies have collectively demonstrated that AL with the explorative selection strategy can iteratively select the representative data from the chemical space to add new information to the model, thereby enhancing its ability to accurately predict the properties of molecules within the space.

Current situation and prospect

Optimizing the integration of advanced machine learning

Research has unequivocally demonstrated that the performance of combined ML models significantly influences the effectiveness of AL.^(p83) Several advanced ML algorithms, such as RL and TL, coupled with automated ML algorithm selection tools, have been seamlessly integrated into AL, yielding highly promising results.^{(p18),(p26)} However, it is worth noting that not all instances of AL combined with advanced ML approaches have found success in the context of drug discovery, as seen with multitask AL.^(p84) Consequently, it is imperative to delve into the effective use of AL incorporating advanced ML algorithms that have proved their worth in other domains to catalyze progress in drug discovery – an area that demands further exploration. Furthermore, as new ML algorithms and automated algorithm selection tools continue to advance at a rapid pace, there is an urgent call to investigate how to efficiently integrate an even wider spectrum of novel and diverse ML algorithms with AL to fully unlock the potential of AL in advancing drug discovery.

Optimizing the development and migration of novel query strategies

As outlined in “The workflow of active learning”, the distinction between explorative and exploitative query strategies is fundamental to the current research landscape. The explorative approach plays a vital role in spatial exploration and enhancing model performance, whereas the exploitative strategy focuses on the selection of molecules with specific desirable properties.^(p7) The significant challenge lies in achieving a delicate balance between advancing model performance and identifying molecules with desired characteristics. Presently, two promising approaches have emerged to address this challenge. First, there is a growing emphasis on the development of novel balanced molecule selection strategies. These strategies aim to identify molecules that not only possess desirable properties but also inject fresh insights into the model.^{(p16),(p48)} Yet, the precise calibration of the balance between explorative and exploitative elements remains a pressing subject for further investigation. Second, researchers are exploring the implementation of distinct query strategies at different stages of the process.^(p18) This involves initially employing an explorative strategy to refine model performance, followed by a transition to an exploitative strategy to pinpoint molecules with sought-after attributes. The optimal timing for this transition for maximizing benefits also requires thorough exploration. Therefore, the ongoing pursuit of innovative strategies that simultaneously enhance model performance and uncover molecules with desirable properties remains a paramount focus in research endeavors.

AL has demonstrated its effectiveness not only in drug discovery but also in various other domains, including material discovery, among others.^{(p85),(p86)} In recent times, novel and advantageous query strategies have emerged in these diverse fields. It is prudent to assess the feasibility of applying these strategies in the drug discovery field or adapting them to suit pharmaceutical research needs. Investigating the potential migration of query strategies from these other domains to expedite drug discovery represents an exciting direction for future research.

Exploring innovative applications of active learning

A comprehensive review of the extensive literature reveals that the primary application of AL primarily revolves around the construction of high-performance ML models and the efficient identification of molecules possessing desirable properties. However, within this landscape, intriguing facets also emerge from certain studies. These investigations showcase that the performance gains and changes in the diversity of generated molecules during the iterative AL process can serve as valuable tools for comparing the information content of molecular characterizations or evaluating the performance of generated models.^{(p26),(p42)} This intriguing insight suggests that the utility of AL may extend beyond its traditional role of model enhancement and the discovery of molecules with desirable properties. As such, there exists a compelling need for continued exploration and the development of novel applications for AL in the field of drug discovery.

Concluding remarks

The advantages of AL-guided data selection align well with the challenges faced in drug discovery, such as the expansion of exploration space and issues with flawed labeled data. Consequently, AL has found extensive applications in the field of drug discovery. However, despite its increasing adoption, there has been a noticeable absence of a comprehensive review of its applications. This paper seeks to address this gap by presenting a relatively comprehensive review of the use of AL across various stages of drug discovery. The systematic review reveals that researchers can harness a variety of query strategies and ML models within the AL workflow to meet their specific research objectives, thus demonstrating the adaptability and versatility of AL. Through this review, our aim is to provide valuable insights and guidance to pharmaceutical researchers, further fostering the integration of AL in the realm of drug development.

Declaration of interests

The authors declare no competing financial interests.

CRediT authorship contribution statement

Lei Wang: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Zhenran Zhou:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Xixi Yang:** Writing – review & editing. **Shaohua Shi:** Writing – review & editing. **Xiangxiang Zeng:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Dongsheng Cao:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization.

Data availability

No data was used for the research described in the article.

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References

- Vamathevan J et al. Applications of machine learning in drug discovery and development. *Nat Rev Drug Discov.* 2019;18:463–477.
- Reda C, Kaufmann E, Delahaye-Duriez A. Machine learning applications in drug development. *Comput Struct Biotechnol J.* 2020;18:241–252.
- Saxe A, Nelli S, Summerfield C. If deep learning is the answer, what is the question? *Nat Rev Neurosci.* 2021;22:55–67.
- Ezzat A, Wu M, Li XL, Kwok CK. Drug-target interaction prediction via class imbalance-aware ensemble learning. *BMC Bioinformatics.* 2016;17:267–276.
- Korkmaz S. Deep learning-based imbalanced data classification for drug discovery. *J Chem Inf Model.* 2020;60:4180–4190.
- Eisenstein M. Active machine learning helps drug hunters tackle biology. *Nat Biotechnol.* 2020;38:512–514.
- Reker D, Schneider G. Active-learning strategies in computer-assisted drug discovery. *Drug Discov Today.* 2015;20:458–465.
- Warmuth MK, Liao J, Rätsch G, Mathieson M, Putta S, Lemmen C. Active learning with support vector machines in the drug discovery process. *J Chem Inf Comp Sci.* 2003;43:667–673.
- Cohn DA, Ghahramani Z, Jordan MI. Active learning with statistical models. *J Artif Intell Res.* 1996;4:129–145.
- MacKay DJ. Information-based objective functions for active data selection. *Neural Comput.* 1992;4:590–604.
- Reker D. Practical considerations for active machine learning in drug discovery. *Drug Discov Today Technol.* 2019;32:73–79.
- Bleicher KH, Bohm HJ, Muller K, Alanine AI. Hit and lead generation: beyond high-throughput screening. *Nat Rev Drug Discov.* 2003;2:369–378.
- Schneider G. Automating drug discovery. *Nat Rev Drug Discov.* 2018;17:97–113.
- Yu J, Li X, Zheng M. Current status of active learning for drug discovery. *Artif Intell Life Sci.* 2021;1 100023.
- Temerinac-Ott M, Naik AW, Murphy RF. Deciding when to stop: efficient experimentation to learn to predict drug-target interactions. *BMC Bioinformatics.* 2015;16:1–10.
- Reker D, Schneider P, Schneider G. Multi-objective active machine learning rapidly improves structure-activity models and reveals new protein-protein interaction inhibitors. *Chem Sci.* 2016;7:3919–3927.
- Ding XY et al. Active learning for drug design: a case study on the plasma exposure of orally administered drugs. *J Med Chem.* 2021;64:16838–16853.
- Gusev F, Gutkin E, Kurnikova MG, Isayev O. Active learning guided drug design lead optimization based on relative binding free energy modeling. *J Chem Inf Model.* 2023;63:583–594.
- Reker D, Schneider P, Schneider G, Brown JB. Active learning for computational chemogenomics. *Future Med Chem.* 2017;9:381–402.
- Fujiwara Y et al. Virtual screening system for finding structurally diverse hits by active learning. *J Chem Inf Model.* 2008;48:930–940.
- Graff DE, Shakhnovich EI, Coley CW. Accelerating high-throughput virtual screening through molecular pool-based active learning. *Chem Sci.* 2021;12:7866–7881.
- Zhang Y, Lee AA. Bayesian semi-supervised learning for uncertainty-calibrated prediction of molecular properties and active learning. *Chem Sci.* 2019;10:8154–8163.
- Cai C et al. Midruglikeness: subdivisional drug-likeness prediction models using active ensemble learning strategies. *Biomolecules.* 2022;13:29.
- Hase F, Roch LM, Kreisbeck C, Aspuru-Guzik A. Phoenix: a bayesian optimizer for chemistry. *ACS Cent Sci.* 2018;4:1134–1145.

25. Seo S-W et al. Accelerating antibody design with active learning. *bioRxiv*. 2022;09(12) 507690.
26. Bengio E, Jain M, Korablyov M, Precup D, Bengio Y. Flow network based generative models for non-iterative diverse candidate generation. *Adv Neural Inf Process Syst*. 2021;34:27381–27394.
27. Kangas JD, Naik AW, Murphy RF. Efficient discovery of responses of proteins to compounds using active learning. *BMC Bioinformatics*. 2014;15:1–11.
28. Buendia R et al. Accurate hit estimation for iterative screening using Venn-ABERS predictors. *J Chem Inf Model*. 2019;59:1230–1237.
29. Li BW, Rangarajan S. Designing compact training sets for data-driven molecular property prediction through optimal exploitation and exploration. *Mol Syst Des Eng*. 2019;4:1048–1057.
30. Reker D, Brown JB. Selection of informative examples in chemogenomic datasets. *Methods Mol Biol*. 2018;1825:369–410.
31. Masoudi-Nejad A, Mousavian Z, Bozorgmehr JH. Drug-target and disease networks: polypharmacology in the post-genomic era. *In Silico Pharmacol*. 2013;1:1–4.
32. Cheng T, Hao M, Takeda T, Bryant SH, Wang Y. Large-scale prediction of drug-target interaction: a data-centric review. *AAPS J*. 2017;19:1264–1275.
33. Sharma A, Rani R. BE-DTI: Ensemble framework for drug target interaction prediction using dimensionality reduction and active learning. *Comput Methods Programs Biomed*. 2018;165:151–162.
34. Rakers C, Najnin RA, Polash AH, Takeda S, Brown JB. Chemogenomic active learning's domain of applicability on small, sparse qHTS matrices: a study using cytochrome P450 and nuclear hormone receptor families. *ChemMedChem*. 2018;13:511–521.
35. Naik AW, Kangas JD, Langmead CJ, Murphy RF. Efficient modeling and active learning discovery of biological responses. *PLoS One*. 2013;8:e83996.
36. Naik AW, Kangas JD, Sullivan DP, Murphy RF. Active machine learning-driven experimentation to determine compound effects on protein patterns. *Elife*. 2016;5:e10047.
37. Chen J, Hou J, Wong KC. Categorical matrix completion with active learning for high-throughput screening. *IEEE/ACM Trans Comput Biol Bioinform*. 2021;18:2261–2270.
38. Sun H, Murphy RF. Evaluation of categorical matrix completion algorithms: toward improved active learning for drug discovery. *Bioinformatics*. 2021;37:3538–3545.
39. Polash AH, Nakano T, Takeda S, Brown JB. Applicability domain of active learning in chemical probe identification: convergence in learning from non-specific compounds and decision rule clarification. *Molecules*. 2019;24:2716.
40. Luo Y, Liu Y, Peng J. Calibrated geometric deep learning improves kinase–drug binding predictions. *Nat Mach Intell*. 2023;5:1390–1401.
41. Lang T, Flachsenberg F, von Luxburg U, Rarey M. Feasibility of active machine learning for multiclass compound classification. *J Chem Inf Model*. 2016;56:12–20.
42. Rodriguez-Perez R, Miljkovic F, Bajorath J. Assessing the information content of structural and protein–ligand interaction representations for the classification of kinase inhibitor binding modes via machine learning and active learning. *J Cheminform*. 2020;12:1–14.
43. Czarnecki WM, Jastrzebski S, Sieradzki I, Podlowska S. Active learning of compounds activity–towards scientifically sound simulation of drug candidates identification. *Proceedings of MLLS: 2nd Workshop on Machine Learning in Life Sciences*:40–51.
44. De Grave K, Ramon J, De Raedt L. Active learning for high throughput screening. *Discovery Science: 11th International Conference, DS 2008, Budapest, Hungary, October 13–16, 2008. Proceedings 11*. Springer; 2008:185–196.
45. Kremer J, Pedersen KS, Igel C. Active learning with support vector machines. *Wires Data Min Knowl*. 2014;4:313–326.
46. Grebner C, Malmerberg E, Shewmaker A, Batista J, Nicholls A, Sadowski J. Virtual screening in the cloud: how big is big enough? *J Chem Inf Model*. 2020;60:4274–4282.
47. Ahmed L et al. Efficient iterative virtual screening with apache spark and conformal prediction. *J Cheminform*. 2018;10:1–8.
48. Yang Y et al. Efficient exploration of chemical space with docking and deep learning. *J Chem Theory Comput*. 2021;17:7106–7119.
49. Gentile F et al. Deep docking: a deep learning platform for augmentation of structure based drug discovery. *ACS Cent Sci*. 2020;6:939–949.
50. Sivula T, Yetukuri L, Kalliokoski T, Kasanen H, Poso A, Pöhner I. Machine learning-boosted docking enables the efficient structure-based virtual screening of giga-scale enumerated chemical libraries. *J Chem Inf Model*. 2023;63:5773–5783.
51. Khalak Y, Tresadern G, Hahn DF, de Groot BL, Gapsys V. Chemical space exploration with active learning and alchemical free energies. *J Chem Theory Comput*. 2022;18:6259–6270.
52. Thompson J et al. Optimizing active learning for free energy calculations. *Artif Intell Life Sci*. 2022;2 100050.
53. Crivelli-Decker JE et al. Machine learning guided AQFEP: a fast & efficient absolute free energy perturbation solution for virtual screening. 2023. <https://doi.org/10.26434/chemrxiv-2023-z3t3b>.
54. Yu J et al. Computing the relative binding affinity of ligands based on a pairwise binding comparison network. *Nat Comput Sci*. 2023;3:860–872.
55. Adeshina YO, Deeds EJ, Karanickolas J. Machine learning classification can reduce false positives in structure-based virtual screening. *Proc Natl Acad Sci USA*. 2020;117:18477–18488.
56. Waszkowycz B, Clark DE, Gancia E. Outstanding challenges in protein–ligand docking and structure-based virtual screening. *Wires Comput Mol Sci*. 2011;1:229–259.
57. Varela R, Walters WP, Goldman BB, Jain AN. Iterative refinement of a binding pocket model: active computational steering of lead optimization. *J Med Chem*. 2012;55:8926–8942.
58. Cao Y, Shen Y. Bayesian active learning for optimization and uncertainty quantification in protein docking. *J Chem Theory Comput*. 2020;16:5334–5347.
59. Hsu DJ, Davidson RB, Sedova A, Glaser J. Tinyfif: a high-throughput binding pose refinement workflow through induced-fit ligand docking. *J Chem Inf Model*. 2023;63:3438–3447.
60. Fusani L, Cabrera AC. Active learning strategies with combine analysis: new tricks for an old dog. *J Comput Aided Mol Des*. 2019;33:287–294.
61. Wang L et al. Reducing false positive rate of docking-based virtual screening by active learning. *Brief Bioinform*. 2023;24:bbac626.
62. de Oliveira C, Leswing K, Feng S, Kanters R, Abel R, Bhat S. FEP protocol builder: optimization of free energy perturbation protocols using active learning. *J Chem Inf Model*. 2023;63:5592–5603.
63. Segler MHS, Kogej T, Tyrchan C, Waller MP. Generating focused molecule libraries for drug discovery with recurrent neural networks. *ACS Cent Sci*. 2018;4:120–131.
64. Zhavoronkov A et al. Deep learning enables rapid identification of potent DDR1 kinase inhibitors. *Nat Biotechnol*. 2019;37:1038–1040.
65. Neil D et al. Exploring deep recurrent models with reinforcement learning for molecule design. *Int Conf Learn Represent*. 2018;6:1–15.
66. Hu W. Exploring local chemical space in de novo molecular generation using multi-agent deep reinforcement learning. *Nat Sci*. 2021;13:412–424.
67. Amabilino S, Pogány P, Pickett SD, Green DVS. Guidelines for recurrent neural network transfer learning-based molecular generation of focused libraries. *J Chem Inf Model*. 2020;60:5699–5713.
68. Iovanac NC, MacKnight R, Savoie BM. Actively searching: inverse design of novel molecules with simultaneously optimized properties. *J Phys Chem a*. 2022;126:333–340.
69. Kusner MJ, Paige B, Hernández-Lobato JM. Grammar variational autoencoder. *International Conference on Machine Learning*. PMLR; 2017:1945–1954.
70. Konze KD et al. Reaction-based enumeration, active learning, and free energy calculations to rapidly explore synthetically tractable chemical space and optimize potency of cyclin-dependent kinase 2 inhibitors. *J Chem Inf Model*. 2019;59:3782–3793.
71. Ghanakota P et al. Combining cloud-based free-energy calculations, synthetically aware enumerations, and goal-directed generative machine learning for rapid large-scale chemical exploration and optimization. *J Chem Inf Model*. 2020;60:4311–4325.
72. Green DVS et al. Bradshaw: a system for automated molecular design. *J Comput Aided Mol Des*. 2020;34:747–765.
73. Yassine R, Makrem M, Farhat F. Active learning and the potential of neural networks accelerate molecular screening for the design of a new molecule effective against SARS-CoV-2. *Biomed Res Int*. 2021;2021:1–14.
74. Desai B et al. Rapid discovery of a novel series of Abl kinase inhibitors by application of an integrated microfluidic synthesis and screening platform. *J Med Chem*. 2013;56:3033–3047.
75. Holub A, Perona P, Burl MC. Entropy-based active learning for object recognition. *2008 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*. IEEE; 2008:1–8.
76. Gal Y, Ghahramani Z. Dropout as a bayesian approximation: representing model uncertainty in deep learning. *International Conference on Machine Learning*. PMLR; 2016:1050–1059.

77. Sushko I et al. Applicability domains for classification problems: benchmarking of distance to models for Ames mutagenicity set. *J Chem Inf Model.* 2010;50:2094–2111.
78. Janet JP, Duan C, Yang T, Nandy A, Kulik HJ. A quantitative uncertainty metric controls error in neural network-driven chemical discovery. *Chem Sci.* 2019;10:7913–7922.
79. Gubaev K, Podryabinkin EV, Shapeev AV. Machine learning of molecular properties: locality and active learning. *J Chem Phys.* 2018;148:24.
80. Hao Z et al. ASGN: an active semi-supervised graph neural network for molecular property prediction. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. 2020; 731–752.
81. Li L, Xiao Y, Ma D, Zheng K. Prevail: pre-trained variational adversarial active learning for molecular property prediction. *2022 IEEE 8th International Conference on Cloud Computing and Intelligent Systems (CCIS).* IEEE; 2022:143–149.
82. Podryabinkin EV, Shapeev AV. Active learning of linearly parametrized interatomic potentials. *Comput Mater Sci.* 2017;140:171–180.
83. Reker D. Active learning for drug discovery and automated data curation. *Artif Intell Drug Discov.* 2020;75:301.
84. Zhang Y, Yang Q. An overview of multi-task learning. *Natl Sci Rev.* 2018;5:30–43.
85. Tian Y et al. Efficient estimation of material property curves and surfaces via active learning. *Phys Rev Mater.* 2021;5 013802.
86. Bassman L et al. Active learning for accelerated design of layered materials. *Npj Comput Mater.* 2018;4:74.